EXPERIMENT 3

Survey Report on Emerging Trends in Low-Code Application Development

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Subject: Low Code Application Development using Zoho Creator
Project: Bank Loan Application & Pre-Screening Management

Aim

To survey the published literature on low-code development, to identify the trends that are shaping it, and to relate them to the Bank Loan Application built on Zoho Creator in this lab series.

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1. Introduction

Low-code development builds applications mainly through visual tools, configuration and reusable components, with hand-written code only where the visual tools are not enough. Business applications, workflow automation, dashboards and internal tools are typical uses. The approach shortens delivery time and lets people who are not professional programmers take part in building software.

2. Low-code, No-code and Pro-code

Table 3.1: Spectrum of development approaches

Approach	Built mainly by	Typical tools	Typical limit
No-code	Business users	Form and workflow designers with no scripting	Limited to what the platform offers
Low-code	Business users and developers together	Visual builders plus a scripting language such as Deluge	Complex logic and scale need extension
Pro-code	Professional developers	General-purpose languages and frameworks	Slowest to deliver; full control

3. Emerging Trends

- **AI-assisted development:** platforms offer prompt-based application generation, AI model builders and AI agents, so that part of the design work starts from a description in plain language.

- **Citizen development with governance:** more of the building is done outside IT departments, which raises the need for roles, environments and review before release.
- **Integration and automation:** applications are connected to other systems through connectors, APIs and workflows; automation is a core use, not an add-on.
- **Pro-code extension:** custom functions, serverless code and APIs are used when configuration is not enough, giving a hybrid of low-code and pro-code.
- **Multi-experience delivery:** the same application is made available on web, mobile and portals.
- **Security and compliance:** role-based access, data sharing rules and audit trails are expected features, especially in finance.

4. Survey Method

This is a desk survey. Published forecasts, scholarly papers and the platform's own documentation were reviewed, and the trends above were drawn from them. No respondent data was collected for this report; a questionnaire for a small class survey is provided in Section 9, with a blank tally table, in case primary data is wanted.

5. Published Forecasts

Gartner's forecast of December 2022, as reported by InfoWorld [1], expected sales of low-code development technologies to reach 44.5 billion US dollars by 2026. It also expected that, by 2026, developers outside formal IT departments would make up at least 80% of the low-code tool user base (60% in 2021), and that low-code tools would account for 75% of new application development (40% in 2021). These are forecasts made in 2022 and should be read as projections, not as measured results.

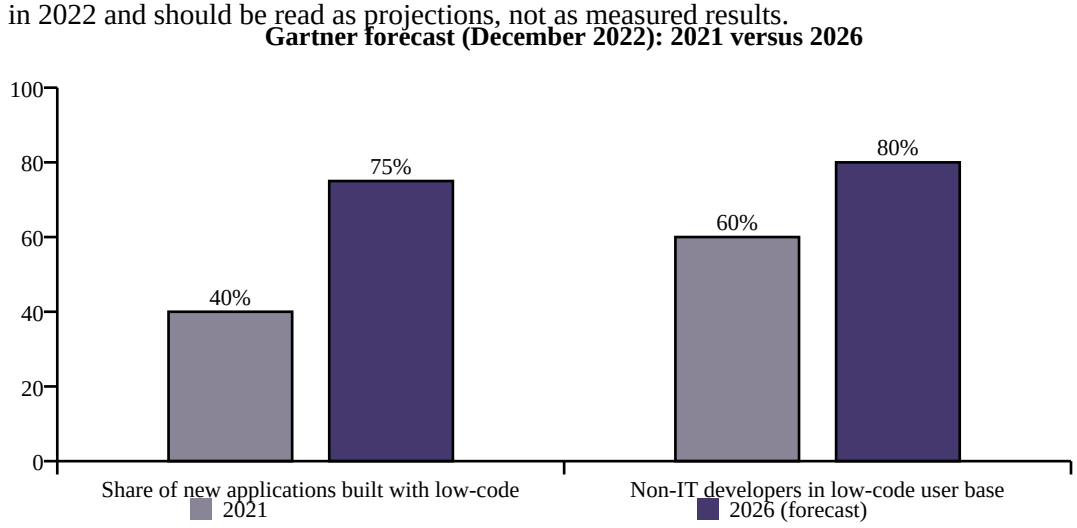


Fig 3.1 — Forecast figures as reported in [1]

6. Popular Platforms

Zoho Creator, Microsoft Power Apps, OutSystems, Mendix, Appian, ServiceNow App Engine and Salesforce Platform tools are commonly named low-code platforms. They differ in focus (business process, enterprise integration, or a wider suite), pricing and the degree of code extension allowed. This lab series uses Zoho Creator throughout.

7. Benefits and Challenges

Table 3.2: Benefits and challenges of low-code development

Benefits	Challenges
Faster delivery of business applications	Platform lock-in; applications are hard to move
Lower coding effort and shorter learning curve	Limits for complex logic or very large data volumes
Business users can take part in building	Governance and shadow-IT risk without roles and review
Built-in forms, reports, workflows and security	Features can be blocked by plan or account settings
Easy integration with other services of the same vendor	Dependence on the vendor’s roadmap and pricing

8. Relevance to the Loan Application

Table 3.3: Trends and the lab experiments

Trend	Where it appears in the lab series
AI-assisted development	Experiment 2 (platform exploration); Experiment 15 (career and future)
Governance	Experiments 8, 11 and 12 (roles, environments, security)
Integration and automation	Experiments 6 and 7 (workflows and Zoho Calendar)
Pro-code extension	Experiment 14 (custom Deluge function)
Limits of low-code	Experiment 13 (case study)

9. Optional Class Questionnaire

If primary data is wanted, the following short questionnaire can be given to classmates. The tally table is left blank on purpose; it is to be filled only with real responses.

1 Have you built an application on a low-code platform? (Yes / No)
2 Which platform? (Zoho Creator / Power Apps / Other / None)
3 How long did your first working form take? (under 1 hour / 1–3 hours / more than 3 hours)
4 Which part was hardest? (forms / reports / workflows / Deluge / deployment)
5 Would you choose low-code for a new internal tool? (Yes / No / Not sure)

Table 3.4: Response tally (to be completed with real data)

Question	Option A	Option B	Option C	Total responses
1				
2				
3				
4				
5				

10. Conclusion

Low-code development is moving towards AI-assisted building, wider participation of non-IT staff, tighter governance and closer links with pro-code extensions. For a regulated setting such as a bank loan process, the practical lessons are to use configuration where possible, to keep logic in short reviewable scripts, and to control access and release carefully.

References

1 A. Venkat, “Low-code development technologies market forecast to hit $44.5 billion by 2026,” InfoWorld, 14 Dec 2022. https://www.infoworld.com/article/2337677/low-code-development-technologies-market-forecast-to-hit-445-billion-by-2026.html

2 A. Bock and U. Frank, “Low-Code Platform,” Business & Information Systems Engineering, vol. 63, no. 6, pp. 733–740, 2021.

3 A. Sahay, A. Indamutsa, D. Di Ruscio and A. Pierantonio, “Supporting the understanding and comparison of low-code development platforms,” in Proc. Euromicro SEAA, 2020, doi: 10.1109/SEAA51224.2020.00036.